

AI Based Smart Dashboard for Electric Vehicles

Narayana Darapaneni¹, Anwesh Reddy Paduri², Sudha B G³, Dilip Kumar Mohapatra⁴, Ghanshyam Ji⁵, Mrudul George⁶, Swathi N⁷

¹ Northwestern University/Great Learning, US

^{2,3} Great Learning, Bangalore, India

⁴⁻⁷ PES University, Bangalore, India
anwesh@greatlearning.in

Abstract. Recently the EV market is blooming all over the world. Due to the concept of smart and safe vehicles, integrated smart dashboard design is getting more importance in these days. An ideal dashboard in electric vehicles should integrate a lot of features like state of charge (SoC) display, range prediction, speed, collision and temperature warnings, charging station proximity etc. Here we are coming up with a smart dashboard using AI techniques which will incorporate all the necessary features. Battery is the source of energy in Electric vehicles. Almost all other battery parameter estimations and other predictions are based on the available battery capacity. Hence a proper estimation of the SoC is important. The non-linear characteristics of the battery make it difficult to estimate SoC properly. Though many studies are happening to improve the design and appeal of the EV dashboards the researches in the direction of accurate SoC estimation are limited. Here we are trying to make a model by adding various features for estimating SoC of battery. To estimate SoC, we will be using various Neural Network algorithms such as ANN, LSTM, and GRU. These algorithms will estimate battery SoC at various ambient temperatures and different battery states. The LSTM approach for SoC estimate is proposed in this research, and the model successfully predicted SoC with low RMSE and MAE. With the help of this fundamental forecast, we want to improve the accuracy of other predictions in our dashboard.

Keywords: SoC (State of charge), SoH (State of Health), Li-ion (Lithium Ion), LiB (Lithium ion Battery), LSTM (Long Short Term Memory), SoE (State of Energy)

1 Introduction

The electric vehicle market is growing all over the world and is starting to go mainstream. Manufactures are competing to make modern stylish dashboards in electric vehicles, since it is one of the most important deciding factors for any electric vehicle buyer. Integrated smart dashboard design is getting more importance in these days. An ideal dashboard has to integrate a lot of features and the dashboard design becomes complex when it has to fit with the main functions of the dashboard. GPS integration of charging stations along with other features is more challenging. Incorporating all

these features without losing the look and appeal is not an easy task. The design of the dashboard for most EVs is still modest and is lacking in the aspect of dimensional design, appeal and other ergonomic aspects. Therefore, the need is to design an efficient and improved smart dashboard for a smooth driving experience. To design a smart dashboard which is a graphical interface, we can use platforms like Android, iOS, Windows etc. An android application is used in this project for the user interface design. Our dashboard will display details such as the calculated time to reach destination, speed, battery charge level, temperature etc. As shown in Fig 1 the dashboard in the proposed model displays the estimated SOC of the battery. Here we are using GRU, RNN, ANN and LSTM to solve various regression and prediction analysis. The precise SoC assessment of the EV battery is the first stage in dashboard development. The SoC is the fundamental battery metric, and it serves as the foundation for practically all other forecasts.

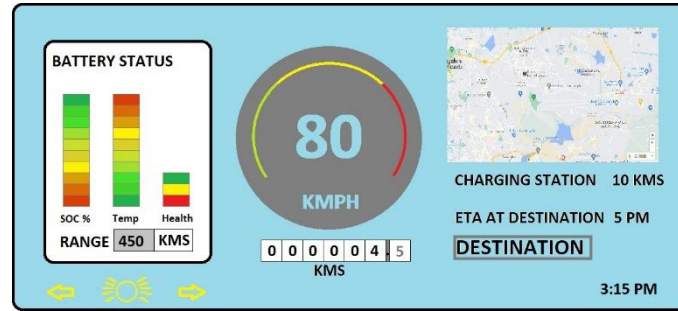


Fig. 1. AI based Smart Dashboard

2 Literature Review

In electric vehicles, the energy comes from the battery. The majority of electric vehicles employ lithium-ion batteries for a variety of reasons. These batteries are well-liked because of their high energy density, extremely low discharge rate, and low cost of maintenance.

A vital battery parameter that shows the battery's usable capacity is state of charge. It is crucial for battery management, so it is also important to estimate. [1]. A battery's state of charge is defined as the ratio between its nominal full capacity and its current available capacity [2] [3]. But the SoC varies with temperature and battery age [4]. Any electric vehicle (EV) driver can benefit from prior knowledge of the battery's State of Charge (SOC), as it helps him determine whether the destination can be reached with the battery's capacity. Without adequate monitoring and upkeep, the battery could experience major problems like overcharging and fire risks [5]. It varies on a number of variables, including the battery's age, temperature, and construction material. [6]. A correct SOC estimation is a difficult endeavor, because of the varied architectures and the selection of hyperparameters [7] [8]. Since the state of charge (SoC) of a battery is

not linearly related to current and temperature, estimating the SoC can be difficult. Direct measurement, book-keeping estimation, and AI techniques can all be used to calculate it [9]. AI methods are the best due to its tremendous learning capabilities. There are a number of SoC estimation methods. Some of them are Coulomb counting method, model-based methods, OCV testing methods, ML and AI methods. Though the Coulomb method is easy, it has some drawbacks [10]. Here the estimated value depends on the initial value of SoC which is not always available. Also, in this method the total accumulated errors are found to be high [11].

OCV (open circuit voltage) are conducted at a stable or idle state of the battery [12]. It requires that the battery be kept in unused state for a long time, usually around 24 hours. This is practically difficult. Model based methods generally use AFA (adaptive filter algorithms) [13] [14]. These algorithms requires a detailed and thorough knowledge about the internal characteristics of the battery. Kalman filter model is built on the assumption that the dynamic model and statistical property of the battery is known in advance and established precisely [15]. Also, the parametrization of the battery requires various tests which are time consuming. [16]. Data driven methods are comparatively simpler when compared with the previous methods. They forecast the output SoC by simply mapping the input values, which are mostly voltage, current, and temperature. They are often used for SoC estimate due of their simplicity. A number of algorithms are there for SoC estimation such as ANN, SVM or extreme learning machine. These methods can be either Machine learning methods or Deep learning methods. Depending upon the cell types and operations we can apply different algorithms to learn battery parameters and SoC of the battery. PFNN algorithm which is an improved version of Fuzzy Neural Network can be applied to lithium battery charge estimation which improves the transient properties of voltage regulation when there is a load change in the battery. The ML methods such as SVR, RF, regression or Gaussian are also used widely due to their simple execution. But ML methods has only 2 layers for input and output [17]. These algorithms take each individual input, process it and give the output. But they don't save any intermediate states or history information. Hence these models cannot be effectively incorporate the complicated electro- chemical and electro-thermal characteristics of the battery.[17] In contrast the DL methods such as CNN and RNN have many computational layers. The efficiency of the deep learning methods can be improved by adding more computational layers [17]. Due to the issue of gradient vanishing, pure RNNs are unable to explain long-term dependencies in the data. The gradient is exponentially back propagated in RNN. But by using a memory cell, LSTM permits the gradient to continue to flow as before. Algorithms like GRU and LSTM (a modified RNN) deal with input variables (current, voltage and temperature) at every time step.

Chennali used LSTM with different layers for SoC estimation and validated the value of SoC at different temperatures. According to his studies, LSTM provides more accurate SoC estimation than other generic algorithms. The degree to which the model has learned information from the training data depends on the SoC prediction accuracy. With the use of gates at various time-steps, the LSTM model gathers data from the training set. The LSTM decides what to remember and what to forget by using input,

forget, and output gates; as a result, it can handle long-term dependencies. LSTM algorithm can collect more information from the training data with the historical time of input for specified time steps.

Since state of battery is a time series data it can be estimated with maximum accuracy with algorithms which have memory to store intermediate battery states. In order to estimate SoC to match with the non-linear features of the battery, the LSTM technique is proposed in this study. Each data point is processed by LSTM, an upgraded version of RNN, which then saves the state with historical information for each layer. The information to be carried forward is stored in the memory cell and it ensures that no information is lost across the layers [18], [19]. Since LSTM is a fully data-driven model, it can be used to estimate the SOC of many battery varieties [20]. Also, the performance accuracy is estimated using MAE and RMSE.

3 Materials And Methods

3.1 Data Collection

We faced real time challenges in collecting the electric vehicles dataset. We got few electric vehicles data set but those were either limited in the data size or limited in the features. Finally, we succeeded in collecting the data from the below sources.

1. Calce [21]: Calce Battery Research Group has provided very good number of datasets related lithium-ion batteries. It has a wide variety of datasets collected at different temperatures with a lot of features. We can use this dataset for all our research after citing the reference.
2. Mendeley [22]: Mendeley data is a free and secured cloud platform where we can use the data for research and publishing by citing the reference of the site. Here we have data with sufficient size and limited number of features where data is captured at different temperatures. The batteries used are Lithium-ion batteries. We decided to go ahead with the above dataset. Our data consists of 13269 rows with 44 features. The data collected is 5 days EV battery (Li-ion) data. The data for Electric Vehicles was collected at different temperatures, different states like Vehicle state charger connected, Vehicle state lock, Vehicle state unlock etc. State of Charge (SoC) is dependent on the ambient temperature and may cause errors in SoC estimation. To address this issue, we collected data for different temperatures like 0deg, 10 deg, 25deg, 40 deg. We have taken 44 features in our model.

3.2 Data Preparation

For our analysis, we have taken the data at 25 0 C room temperature. The data is stored in an excel file which has around 13269 rows. The data had few columns having more than 80 percent of the data as null. We analyzed the respective domains of those columns and discussed that with the field experts and then understood that there is no significance of those features. Hence, we removed those features which had many nulls.

We have cleaned the data and removed the outliers using the IQR method. We have imputed the nulls in the data. Since we had a categorical variable, we applied the get dummies function and encoded it. The correlation between different features have been plotted using a heat map. The most important features such as current, voltage, and temperature for SoC estimation have been identified and plotted. In our analysis we noticed that the voltage is negatively skewed for 30 bins. Since it is negatively skewed the mean < median < mode.

3.3 Data Exploration

- 1) *Scatter Plot*: As shown in the diagram fig 3, shows us the relation between the quantitative variables current and the voltage. As we see in the graph most of the voltage is distributed around +10 and +13 current. Very few voltage values are distributed between from -15 to +10 current. We can observe with the color codes how the temperature is distributed. As we observed there is a positive correlation between the Voltage and Current of the battery at different temperatures.

We also observed that there is a positive correlation between Voltage and Current of the battery using a scatter plot and also, we observed that there is a positive correlation between the voltage, current and temperature.

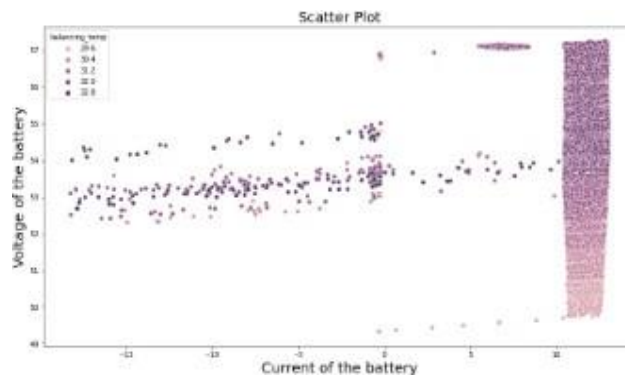


Fig. 2. Scatter Plot

3.4 Data Visualization

1. *Current and Voltage Plot*: Current and the voltage are directly proportional. When there is an increase in the voltage current is also increased. In fig 3 Current remains constant at certain threshold.

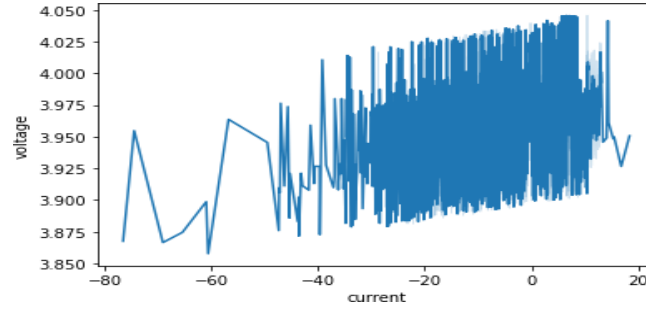


Fig. 3. Current versus Voltage

2. *Measured SoC versus Voltage:* As we can see in the graph 4 how the measured SoC is distributed across the voltage. Voltage and display SoC are linearly related.

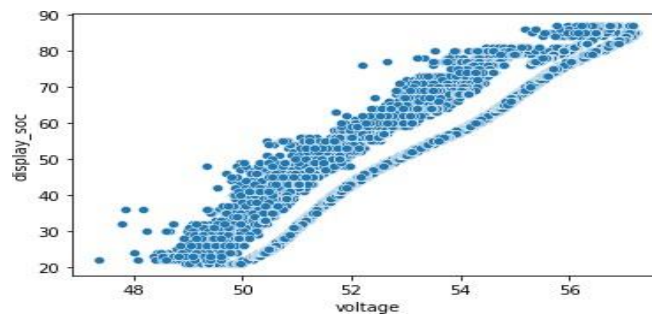


Fig. 4. Measured SoC versus Voltage

3. *Measured SoC versus State:* We can see in the graph 6 how the measured SoC is distributed across the voltage. Voltage and display SoC are linearly related.

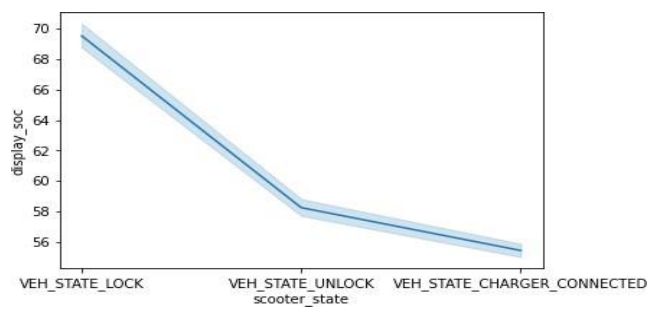


Fig. 5. Measured SoC versus State

4. *Actual and Prediction data for ANN:* Actual and prediction have lots of differences as shown in fig 6.

ANN loss refers to the difference between the predicted output of the model which is shown in the red line and the actual target value which is shown in the blue line. We have used MAE as our loss function.

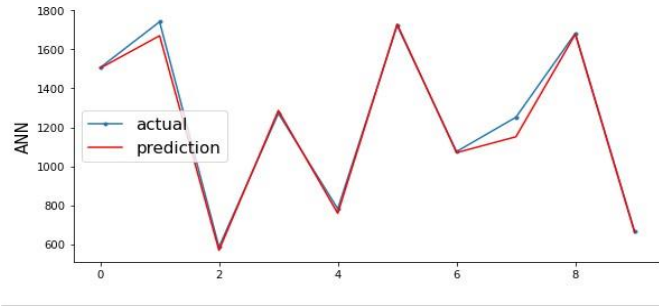


Fig. 6. Actual and Prediction data for ANN

5. *Actual and Prediction data for LSTM:* Calculation of the actual and prediction loss in LSTMs is similar to that in ANNs, but the architecture of LSTMs allows them to handle sequential data and capture long-term dependencies. We have used MAE as our loss function. Actual and prediction loss are almost similar as shown in 7.

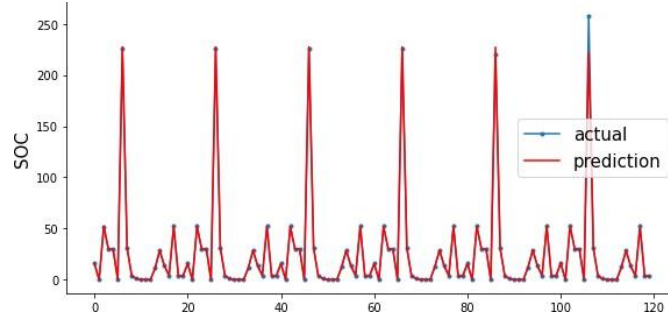


Fig. 7. Actual and Prediction data for LSTM

6. *Actual and Prediction data for GRU:* Calculation of the actual and prediction in GRUs is like that in LSTMs and ANNs, but the architecture of GRUs allows it to handle sequential data and capture long-term dependencies in a computationally efficient manner compared to LSTMs.

We have used MAE as our loss function. Actual and prediction loss have slight differences as shown in the fig 8.

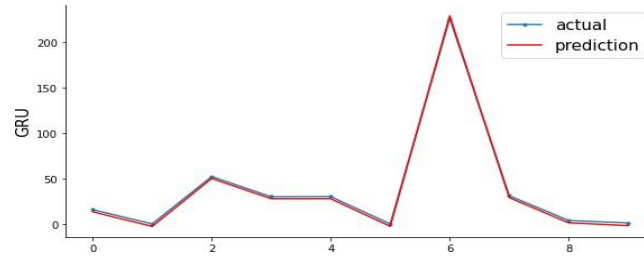


Fig. 8. Actual and Prediction data for GRU

7. *Train and Test Loss of LSTM*: The train and test loss of an LSTM model refers to the measurement of the error in the output produced by the model on the training data (train loss) and on unseen data (test loss). As shown in fig 9, the model seems to fit correctly.

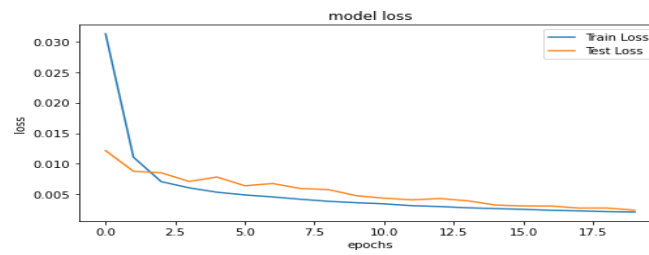


Fig. 9. Train and Test Loss of LSTM

8. *Train and Test Loss of GRU*: A GRU model's "train and test loss" refers to the measurement of the output error on training data (train loss) and on unobserved data (test loss).

As shown in fig 10 we monitor the train and test loss during the process, it seems to be fit correctly. The train and test loss in GRUs is slightly more than that in LSTMs.

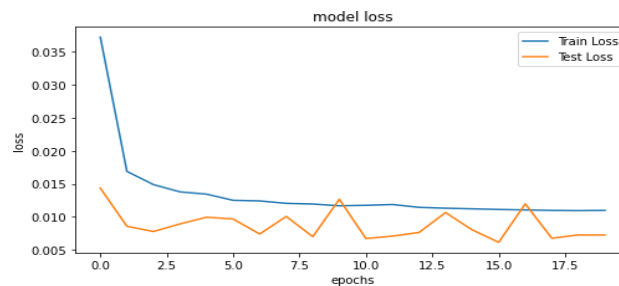


Fig. 10. Train and Test Loss of GRU

4 Discussion And Conclusion

For smart dashboard SoC estimation is important as all other data is dependent on SoC percentage value. So, we started our research on SoC estimation.

Dataset was collected from various sources such as Calce dataset and Mendeley dataset. Our experiment included additional features such as vehicle state, total regenerative current etc. as input to ANN, GRU and LSTM models to estimate SoC more precisely. As part of data exploration some of our observations are:

- Current increases with respect to voltage of the LiB initially for 20 percent of the voltage and then remains steady.
- We found that battery SoC decreases fast when vehicle is in unlocked state.
- SoC percentage decreases as temperature increases.

By fitting our data into a hyperplane that can describe the link between the different variables, we are trying to determine the effectiveness of various algorithms for predicting SOC in this case. Applying various memory-integrated algorithms allows us to do this. These algorithms use similar training techniques and base their output on previous input. All of the network's layers have the same parameters. Because of the size of the dataset, LSTM is preferable here. GRU, however, is quicker for smaller datasets. They take longer to train and have a lower convergence rate. LSTMs learn more quickly than GRUs since they have more parameters.

In our investigation, the LSTM reaches the least loss point and the best accuracy after running the model for 26 epochs. The loss curve remains unchanged after this specific point, demonstrating the highest accuracy.

We found from the experimental results that, as shown in table 1, out of all the models tested, LSTM gives SoC percentage with highest accuracy. It has the lowest MAE value of 0.8450. Hence, we selected LSTM algorithm to use in our smart dashboard for display of SoC.

Table 1. I COMPARISON OF LOSS WITH DIFFERENT MODELS

Models		
Model Name	RMSE	MAE
ANN	35.2414	20.3774
LSTM	4.0595	0.8450
GRU	5.0522	2.5490

As shown in fig 11 ANN has higher loss when compared with the LSTM, GRU

also has slightly higher loss when compared with LSTM. So, we have reached the conclusion that for SoC models LSTM performs better and gives the best result with the loss measure MAE.

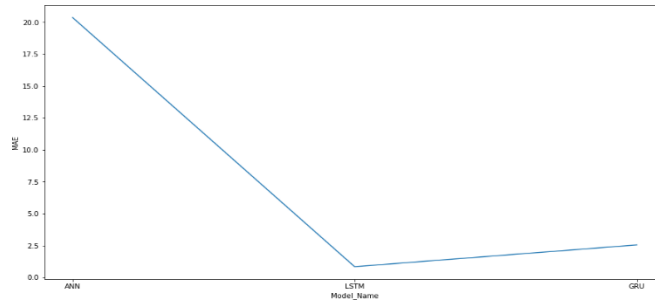


Fig. 11. Line graph for loss for three different models

Therefore, we are developing a dashboard that will include Current, Voltage, Temperature and Battery State to estimate the SoC of the battery using a variety of neural network methods, including ANN, GRU, and LSTM. But we discovered that LSTM combined with MAE yields the best outcome.

5 References

1. M. Trivedi, R. Kakkar, R. Gupta, S. Agrawal, S. Tanwar, V.-C. Niculescu, M. S. Raboaca, F. Alqahtani, A. Saad, and A. Tolba, "Blockchain and deep learning-based fault state of charge and state of energy estimation for lithium-ion batteries based on a long short-term memory neural network. detection framework for electric vehicles," *Mathematics*, vol. 10, no. 19, p. 3626, 2022.
2. W. Wang, X. Wang, C. Xiang, C. Wei, and Y. Zhao, "Unscented kalman filter-based battery soc estimation and peak power prediction method for power distribution of hybrid electric vehicles," *Ieee Access*, vol. 6, pp. 35 957–35 965, 2018.
3. X. Wu, X. Li, and J. Du, "State of charge estimation of lithium-ion batteries over wide temperature range using unscented kalman filter," *Ieee Access*, vol. 6, pp. 41 993–42 003, 2018.
4. M. H. Lipu, M. Hannan, A. Hussain, A. Ayob, M. H. Saad, T. F. Karim, and D. N. How, "Data-driven state of charge estimation of lithium-ion batteries: Algorithms, implementation factors, limitations and future trends," *Journal of Cleaner Production*, vol. 277, p. 124110, 2020.
5. A. J. Ilott, M. Mohammadi, C. M. Schauerma, M. J. Ganter, and A. Jerschow, "Rechargeable lithium-ion cell state of charge and defect detection by in-situ inside-out magnetic resonance imaging," *Nature communications*, vol. 9, no. 1, p. 1776, 2018.
6. J. Y. Yong, V. K. Ramachandaramurthy, K. M. Tan, and N. Mithulananthan, "A review on the state-of-the-art technologies of electric vehicle, its impacts and prospects," *Renewable and sustainable energy reviews*, vol. 49, pp. 365–385, 2015.
7. M. A. Hannan, D. N. How, M. B. Mansor, M. S. H. Lipu, P. J. Ker, and
8. K. M. Muttaqi, "State-of-charge estimation of li-ion battery using gated recurrent unit with one-cycle learning rate policy," *IEEE Transactions on Industry Applications*, vol. 57, no. 3, pp. 2964–2971, 2021.

9. D. N. How, M. A. Hannan, M. S. H. Lipu, K. S. Sahari, P. J. Ker, and K. M. Muttaqi, "State-of-charge estimation of li-ion battery in electric vehicles: A deep neural network approach," *IEEE Transactions on Industry Applications*, vol. 56, no. 5, pp. 5565–5574, 2020.
10. M. H. Lipu, M. Hannan, T. F. Karim, A. Hussain, M. H. M. Saad,
11. A. Ayob, M. S. Miah, and T. I. Mahlia, "Intelligent algorithms and control strategies for battery management system in electric vehicles: Progress, challenges and future outlook," *Journal of Cleaner Production*, vol. 292, p. 126044, 2021.
12. Z. Li, J. Huang, B. Y. Liaw, and J. Zhang, "On state-of-charge determination for lithium-ion batteries," *Journal of Power Sources*, vol. 348, pp. 281–301, 2017.
13. X. Hu, F. Feng, K. Liu, L. Zhang, J. Xie, and B. Liu, "State estimation for advanced battery management: Key challenges and future trends," *Renewable and Sustainable Energy Reviews*, vol. 114, p. 109334, 2019.
14. F.-J. Lin, M.-S. Huang, P.-Y. Yeh, H.-C. Tsai, and C.-H. Kuan, "Dsp-based probabilistic fuzzy neural network control for li-ion battery charger," *IEEE Transactions on Power Electronics*, vol. 27, no. 8, pp. 3782–3794, 2012.
15. J. C. A. Anton, P. J. G. Nieto, C. B. Viejo, and J. A. V. Vilán, "Support vector machines used to estimate the battery state of charge," *IEEE Transactions on power electronics*, vol. 28, no. 12, pp. 5919–5926, 2013.
16. M. S. H. Lipu, M. A. Hannan, A. Hussain, M. H. Saad, A. Ayob, and M. N. Uddin, "Extreme learning machine model for state-of-charge estimation of lithium-ion battery using gravitational search algorithm," *IEEE Transactions on Industry Applications*, vol. 55, no. 4, pp. 4225–4234, 2019.
17. G. S. Misyris, D. I. Doukas, T. A. Papadopoulos, D. P. Labridis, and
18. G. Agelidis, "State-of-charge estimation for li-ion batteries: A more accurate hybrid approach," *IEEE Transactions on Energy Conversion*, vol. 34, no. 1, pp. 109–119, 2018.
19. R. Xiong, J. Cao, Q. Yu, H. He, and F. Sun, "Critical review on the battery state of charge estimation methods for electric vehicles," *Ieee Access*, vol. 6, pp. 1832–1843, 2017.
20. Y. Liu, G. Zhao, and X. Peng, "Deep learning prognostics for lithium-ion battery based on ensemble long short-term memory networks," *IEEE Access*, vol. 7, pp. 155 130–155 142, 2019.
21. X. Song, F. Yang, D. Wang, and K.-L. Tsui, "Combined cnn-lstm network for state-of-charge estimation of lithium-ion batteries," *Ieee Access*, vol. 7, pp. 88 894–88 902, 2019.
22. "Combined state of charge and state of health estimation over lithium-ion battery cell cycle lifespan for electric vehicles," *Journal of Power Sources*, vol. 273, pp. 793–803, 2015.
23. M. Bercibar, I. Gandiaga, I. Villarreal, N. Omar, J. Van Mierlo, and
24. P. Van den Bossche, "Critical review of state of health estimation methods of li-ion batteries for real applications," *Renewable and Sustainable Energy Reviews*, vol. 56, pp. 572–587, 2016.
25. calcel, "Calce battery research group."
26. V. C. N. M. S. M. Kollmeyer, Philip, "Lg 18650hg2 li-ion battery data and example deep neural network xev soc estimator script."